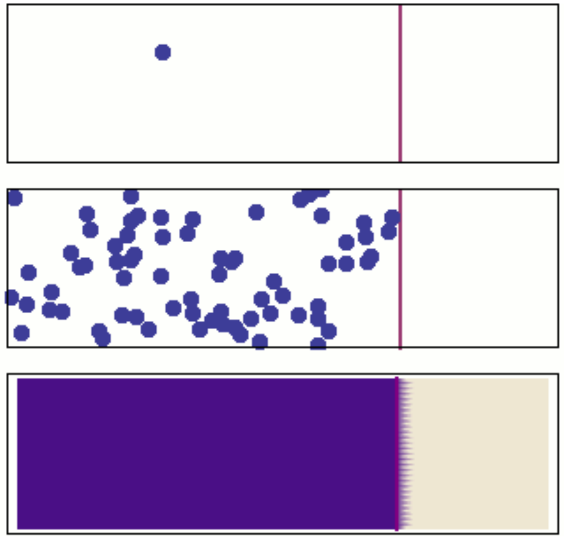
Lecture Unit 3.4

The diffusion equation

What is diffusion?

- Diffusion is the net movement of a quantity from a region of high concentration to a region of lower concentration
- Diffusion is caused by the random motions of molecules in liquids and gases e.g. spreading of a dye in a liquid, heat transfer
- The same equation describes other important processes in physical sciences, life sciences and social sciences



https://en.wikipedia.org/wiki/Diffusion

The 1-D diffusion equation

- The diffusion of a scalar field $u(x,t)$ is described by the diffusion equation:

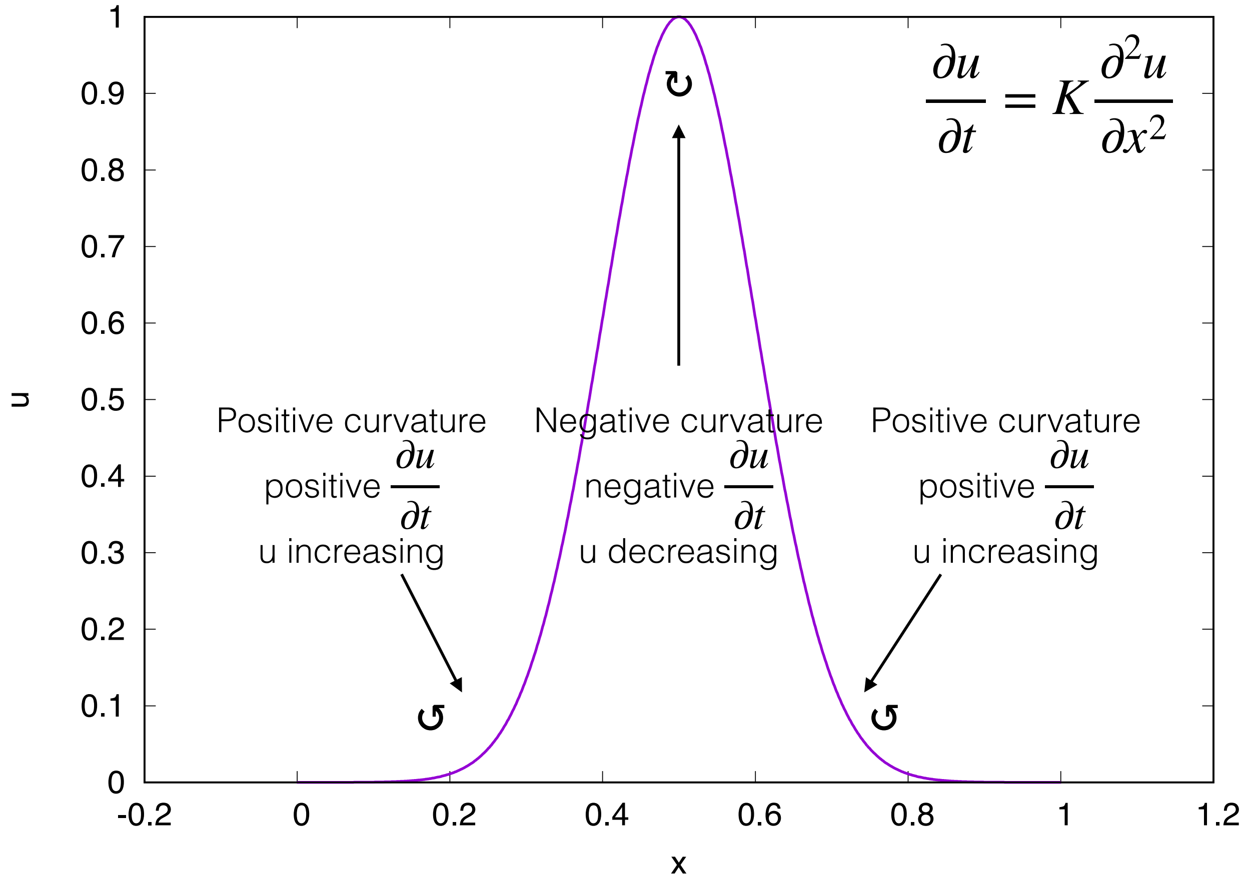
$$\frac{\partial u}{\partial t} = K \frac{\partial^2 u}{\partial x^2}$$

where K is a constant

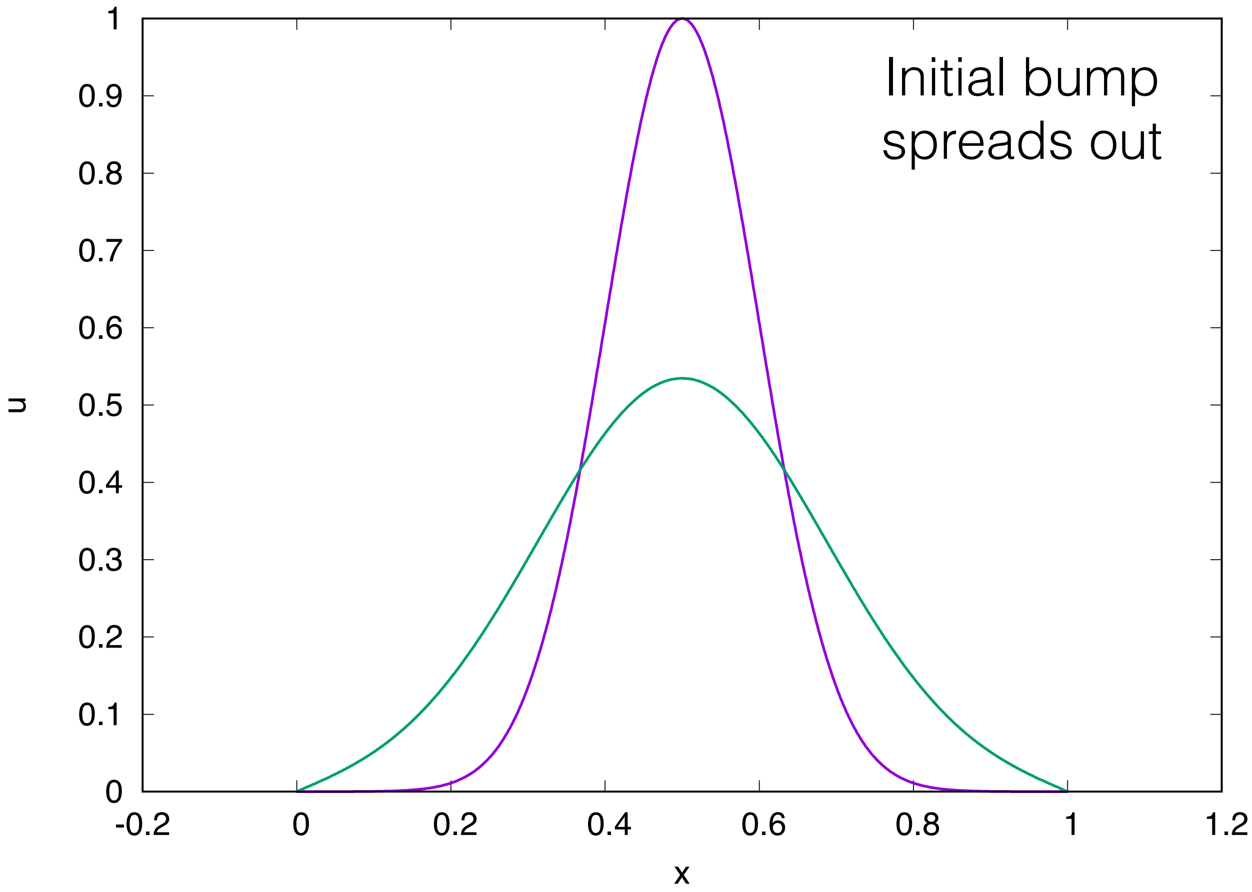
- In this equation the right hand side involves a second derivative of $u(x,t)$ with respect to x

- The second derivative is the gradient of the gradient which measures curvature

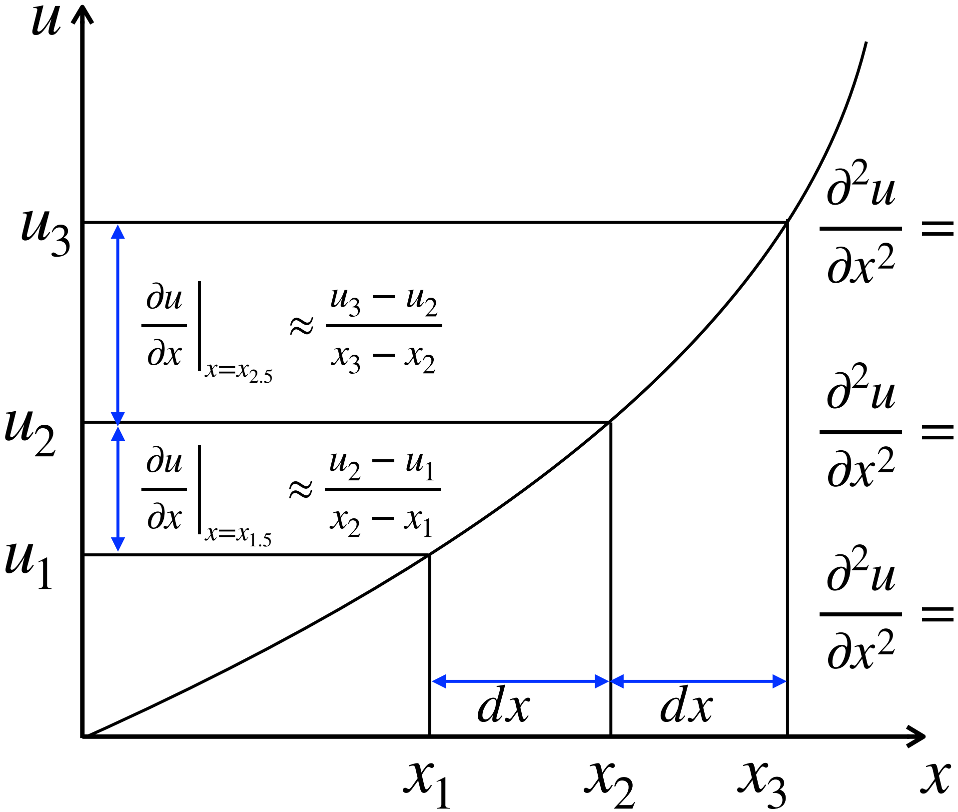
The 1-D diffusion equation



The 1-D diffusion equation



What is the gradient of the gradient?



$$\frac{\partial^2 u}{\partial x^2} = \frac{\left.\frac{\partial u}{\partial x}\right|_{x=x_{2.5}} - \left.\frac{\partial u}{\partial x}\right|_{x=x_{1.5}}}{dx}$$

$$\frac{\partial^2 u}{\partial x^2} = \frac{\frac{u_3 - u_2}{dx} - \frac{u_2 - u_1}{dx}}{dx}$$

$$\frac{\partial^2 u}{\partial x^2} = \frac{u_3 - 2u_2 + u_1}{(dx)^2}$$

Finite differences

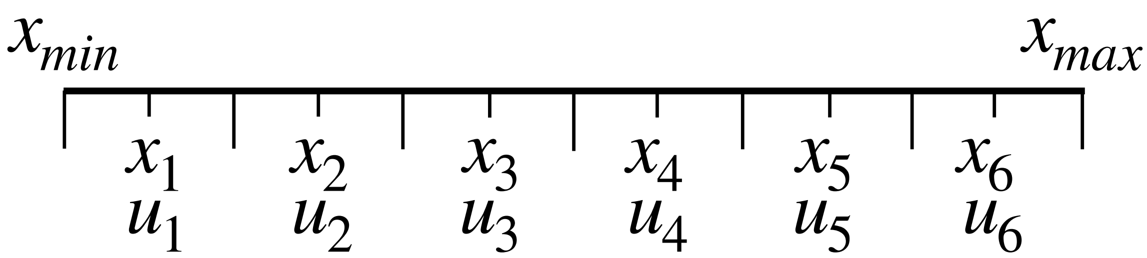
We can use the same grid that we used for the advection equation with our new finite difference expression:

Value on right

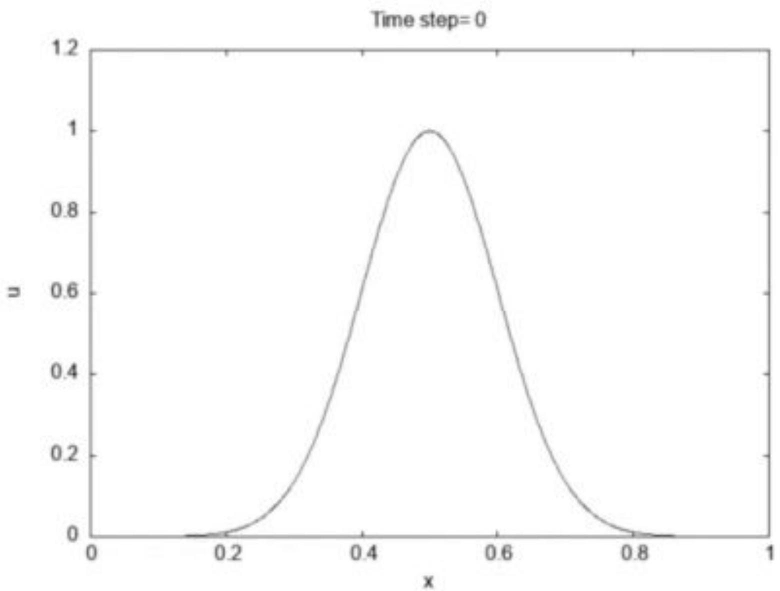
Value on left

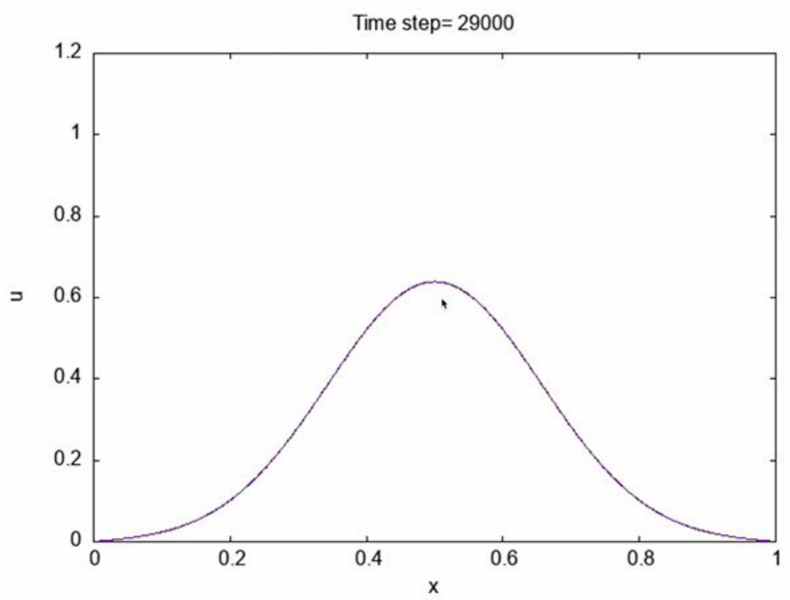
$$\left.\frac{\partial^2 u}{\partial x^2}\right|_{x=x_i} \approx \frac{u_{i+1} - 2u_i + u_{i-1}}{(dx)^2}$$

We will need to use two boundary values this time



Finite differences





The 2-D diffusion equation

- The diffusion of a scalar field $u(x, y, t)$ is described by the diffusion equation:

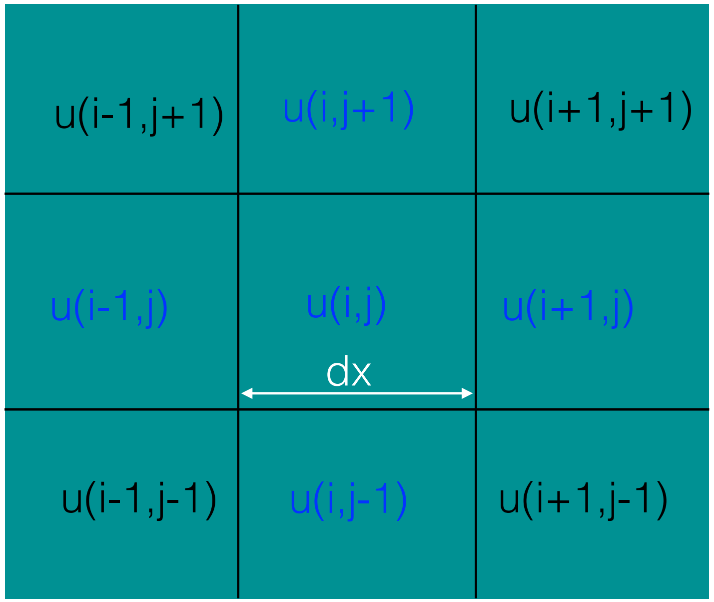
$$\frac{\partial u}{\partial t} = K\left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}\right) = K\nabla^2 u$$

where K is a constant

- In this equation the right hand side involves two second derivatives of $u(x, y, t)$ with respect to x and y

The 2-D diffusion equation

Five point stencil:



$$\nabla^2 u = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}$$

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del2u =(u(i+1,j) - 2*u(i,j) + u(i-1,j))/(dx*dx) +
       (u(i,j+1) - 2*u(i,j) + u(i,j-1))/(dy*dy)
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Unit Summary

- The diffusion equation is a partial differential equation containing a time derivative and one or more second order spatial derivatives
- Second order spatial derivatives can also be calculated using a finite difference approximation
- We can calculate a numerical solution using time stepping and a finite difference stencil like the method used for the advection equation